

Treadmill Offset Calibration and Modeling Report

A Methodological Overview for Thesis Documentation

1. Objective and Problem Statement

Ground reaction forces (GRFs) collected from an instrumented treadmill often exhibit baseline offsets even when the treadmill is empty. These offsets are not static; they fluctuate depending on the mechanical state of the treadmill, specifically the belt speed and the incline (slope) of the machine. Failing to correct for these dynamic offsets propagates errors into the Inverse Dynamics (ID) pipeline, resulting in inaccurate joint moment and joint power calculations. The objective of this methodology was to quantify these empty-treadmill offsets across a wide range of operational conditions and develop a robust, dynamic correction model to tare the force plates accurately during experimental trials.

2. Data Collection and Pre-processing

Calibration data was collected by running the empty treadmill across a dense grid of conditions: 15 different speeds (ranging from 0.2 mph to 1.5 mph) at 6 different slopes (0%, 2.1%, 2.5%, 3.1%, 3.5%, 4.0%, and 4.5%). For each condition, a .mot file was recorded containing the forces for Plate 4 (Left) and Plate 5 (Right). A data parsing script was developed to iterate through all 101 recorded .mot files. For each file, the median and mean values were computed over time for the six relevant force channels: Fx (medial-lateral shear), Fy (vertical), and Fz (anterior-posterior shear). The median was selected as the primary metric to ensure robustness against transient mechanical vibrations or noise spikes.

3. Linear Modeling and Analysis

An initial hypothesis posited that the relationship between the machine's offset, speed, and slope could be described using a Multiple Linear Regression model ($\text{Offset} = B_0 + B_1 \cdot \text{Speed} + B_2 \cdot \text{Slope}$). We fit independent OLS regression models for each force axis. The analysis yielded the following key insights concerning the coordinate system:

- X-Axis (Medial-Lateral Shear): Displayed a very strong linear relationship with speed and slope, with R-squared values reaching approximately 0.83.
- Y-Axis (Vertical Force): Exhibited a moderate linear relationship, with R-squared values ranging between 0.30 and 0.54.
- Z-Axis (Anterior-Posterior Shear): Showed an extremely poor linear fit, with R-squared values falling below 0.02. A simple linear plane could not explain the variance in these specific offsets.

Because the linear models failed to reliably predict the vertical and anterior-posterior shear forces, a unified linear approach was rejected in favor of a non-linear, multidimensional model.

4. Final Model Selection: 2D Grid Interpolation

Given the density of the calibration grid, the optimal solution implemented was a 2D Grid Interpolation (Bivariate Spline / Lookup Table) approach. Six completely independent interpolator models were constructed (one for each force axis per plate) using Scipy's LinearNDInterpolator. This approach avoids forcing a specific mathematical formula onto the complex mechanical behavior of the treadmill. Instead, it constructs a smooth, continuous 2D surface mathematically tying the empirical calibration points together. When querying the model for an experimental condition (e.g., Speed = 1.25 mph, Slope = 3.3%), the interpolator procedurally estimates a highly accurate offset by weighting the nearest known calibration medians.

5. Implementation and Pipeline Integration

A reusable Python class, `TreadmillOffsetCorrector`, was developed to encapsulate this logic. Integrated directly into the ID computing pipeline, this module dynamically tares the force plate data array by subtracting the interpolated baseline offsets before the data is handed to the OpenSim Inverse Dynamics solver. This procedural correction ensures that the resultant joint power calculations are untainted by machine-specific baseline shifts.

Figure 1: 3D Visualization of Speed, Slope, and Force Offset with Regression Plane

